

Beyond CNNs: Leveraging Swin-Transformers for Precise Visual Saliency Prediction

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Abstract—Accurately predicting visual saliency remains a critical challenge in computer vision, essential for simulating the selective attention mechanisms of the Human Visual System (HVS). While Convolutional Neural Networks (CNNs) have driven significant improvements in human attention prediction, Vision Transformers offer a powerful alternative for capturing global contextual dependencies. In this report, we propose an end-to-end deep learning architecture based on a Swin-Transformer (Swin-T) encoder paired with a skip-layer convolutional decoder. Our model captures hierarchical saliency information, merging deep, coarse global features with shallow, fine-grained local responses. To optimize learning, we introduce a custom Hybrid Loss function that balances Kullback-Leibler (KL) divergence with the Pearson Correlation Coefficient (CC). Furthermore, we implement a robust inference pipeline featuring Multi-Scale Test-Time Augmentation (TTA). Experimental evaluations on SALICON dataset, demonstrate that our method achieves highly competitive results, including a CC of 0.9037 and a SIM of 0.7903, positioning it as a robust solution for state-of-the-art visual attention prediction.

Index Terms—Unsupervised Learning, Encoder/Decoder, Neural Networks, Convolutional Neural Networks, Vision Transformers.

I. INTRODUCTION

Humans possess an astonishing ability to quickly direct visual attention to specific regions of a scene rather than processing it in its entirety. Simulating this selective attention mechanism, commonly known as visual saliency prediction, remains a foundational challenge in computer vision. Accurate computational models of attention not only provide deep insights into the Human Visual System (HVS) but also serve as critical precursors for complex downstream tasks, ranging from automated image cropping and object recognition to advanced video understanding.

In recent years, Convolutional Neural Networks (CNNs) have driven a paradigm shift in bottom-up saliency detection by automatically extracting hierarchical feature representations directly from raw pixels. While traditional deep architectures have achieved remarkable success, they inherently rely on localized receptive fields. Capturing the overarching global context of an image often requires aggressively stacking convolutional layers or employing resource-heavy multi-scale input streams. These approaches can lead to redundant computations and frequently struggle to perfectly integrate coarse global semantics with fine-grained local spatial details.

To address these structural limitations, this report presents an end-to-end deep learning architecture that leverages the global context-awareness of Vision Transformers combined with the precise spatial localization of CNNs. We tackle the problem of continuous human eye fixation prediction by proposing a hybrid encoder-decoder model. Our approach utilizes a pre-trained Swin-Tiny (Swin-T) transformer [1] to extract hierarchical, multi-scale features, seamlessly bridging them into a convolutional decoder via skip connections. To ensure the predicted saliency maps accurately reflect both the underlying probability distribution and the structural layout of ground truth fixations, we introduce a custom hybrid loss function. This framework demonstrates high efficiency and yields highly competitive performance, making it readily applicable for integration into practical computer vision pipelines that require fast, attention-driven image analysis.

The main contributions of this work are summarized as follows:

- **Transformer-Based Architecture:** We design a skip-layer network featuring a Swin-T encoder and a CNN decoder augmented with spatial dropout. This effectively merges coarse global attention contexts with fine-grained local responses within a single unified stream.
- **Hybrid Loss Optimization:** We introduce the KLC-CHybridLoss, which directly optimizes the network by treating statistical distribution overlap (via KL Divergence) and linear structural correlation (via the Pearson Correlation Coefficient) with equal priority.
- **Advanced Inference Pipeline:** We implement a robust inference strategy utilizing Multi-Scale Test-Time Augmentation (TTA) alongside a dynamic grid search for optimal Gaussian blur parameters, maximizing the fidelity of predictions on unseen data.

This report is structured as follows. In Section II, we review the related work and state of the art in visual saliency prediction. Section III details the proposed modular processing pipeline, while the data pre-processing and formatting criteria are presented in Section IV. The custom deep learning framework and the optimization schedule are described in Section V. Section VI provides a thorough quantitative and qualitative evaluation of the experimental results. Finally, Section VII concludes the report with a discussion on future extensions and practical insights.

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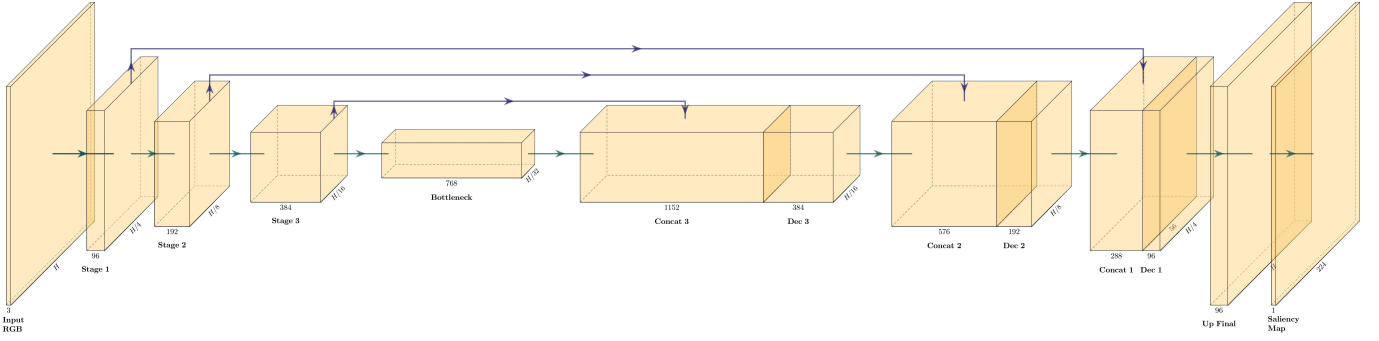


Fig. 1: Architecture of the proposed model. The diagram illustrates the hierarchical feature extraction via the Swin-Transformer encoder, the skip-connection-based convolutional decoder, and the progressive reconstruction of the saliency map at original input resolution.

II. RELATED WORK

A. Traditional and Early CNN-Based Saliency

The foundational models of visual attention prediction relied heavily on hand-crafted features, such as color, intensity, and edge orientation, to compute center-surround contrast. While these heuristic models provided valuable insights into the human attention system, their representational capacity was fundamentally limited. The advent of Convolutional Neural Networks (CNNs) revolutionized saliency detection by enabling the automatic extraction of hierarchical features. Early deep learning architectures, such as single-stream networks (e.g., DeepFix [2]), demonstrated significant performance leaps over traditional methods. By completely replacing hand-crafted features with an end-to-end learning approach, DeepFix successfully captured both local details and global image contexts, establishing new state-of-the-art results on challenging datasets like MIT300 and CAT2000.

B. Skip-Layer Networks and their Limitations

To address the inefficiencies of multi-stream models, skip-layer architectures were proposed. A prominent example is the Deep Visual Attention (DVA) model proposed by Wang and Shen [3]. Instead of utilizing multiple input scales, DVA extracts multi-scale saliency information intrinsically by connecting deep, coarse layers (capturing global context) with shallow, fine layers (capturing local details) within a single VGG16 network.

While the DVA model and similar skip-layer CNN architectures successfully integrate multi-level features, they are still inherently constrained by the fundamental mechanics of convolution. Standard CNNs rely on localized receptive fields; to achieve a truly global understanding of the image context, they require aggressively stacked layers and heavy spatial downsampling. This aggressive downsampling often destroys the fine-grained spatial resolution required for generating precise, continuous fixation maps.

C. Extending the State of the Art with Vision Transformers

These limitations of CNN-based global feature extraction are precisely what we addressed within our project. We

extend the state of the art by replacing the traditional CNN encoder with a Vision Transformer, specifically the Swin-T architecture [1]. Unlike CNNs, Transformers compute self-attention, inherently capturing long-range global dependencies and semantic contexts without requiring excessively deep convolutional cascades. Furthermore, previous models often treat saliency prediction as a simple pixel-wise classification task optimized via standard cross-entropy. Our work improves upon this by introducing a custom KLCCHybridLoss function. This directly minimizes the distribution divergence (Kullback-Leibler) while simultaneously maximizing the structural layout (Pearson Correlation Coefficient) of the predictions.

III. PROCESSING PIPELINE

The architecture is built upon a modular framework, encompassing data preprocessing, hierarchical feature extraction, hybrid optimization, and an advanced inference mechanism. A conceptual overview of the system's blocks and their interrelations is illustrated in Fig. 1.

A. Encoder (Swin-Transformer)

The encoder leverages a Swin-T model, pre-trained on ImageNet [4], to extract multi-scale feature representations. To interface properly with the convolutional decoder, the native Swin-T outputs formatted as (Batch, Height, Width, Channels) are permuted to a standard (Batch, Channels, Height, Width) format. Features are extracted at four distinct stages:

- **Stage 1:** Produces initial features with 96 channels at 1/4th of the original spatial resolution.
- **Stage 2:** Generates features with 192 channels at 1/8th of the spatial resolution.
- **Stage 3:** Outputs features with 384 channels at 1/16th of the spatial resolution.
- **Stage 4 (Bottleneck):** Extracts the deepest global features with 768 channels at 1/32nd of the spatial resolution.

B. Decoder (Convolutional with Skip Connections)

The decoder is responsible for reconstructing the saliency map by progressively upsampling the bottleneck features and fusing them with the corresponding encoder features via skip connections.

- **Upsampling:** The network uses bilinear interpolation with a scale factor of 2 to enlarge the feature maps before merging them with the encoder stages.
- **Skip Connections:** The upsampled features are concatenated with the outputs from Stage 3, Stage 2, and Stage 1, respectively.
- **Decoder Blocks:** After concatenation, the merged features pass through a custom decoder block consisting of a 3x3 Convolution, Batch Normalization, and a ReLU activation. This is immediately followed by a Spatial Dropout layer (with a probability of $p=0.2$) which randomly drops 20% of the channels to prevent overfitting. The block concludes with another 3x3 Convolution, Batch Normalization, and ReLU activation.
- **Final Output Layer:** After the three decoding stages, a final bilinear upsampling with a scale factor of 4 restores the feature map to the original input resolution. A 1x1 convolution followed by a Sigmoid activation function is then applied to generate the continuous 2D saliency map.

IV. SIGNALS AND FEATURES

To train and evaluate the proposed architecture, we utilize the benchmark SALICON dataset [5], which represents a widely accepted reference for visual attention prediction. The input signals consist of natural scenes formatted as RGB images, while the corresponding target ground truths are provided as continuous 2D grayscale distributions representing human eye fixation maps.

A. Data Pre-processing and Formatting

Prior to network ingestion, both the input stimuli and the ground truth maps undergo a strict pre-processing pipeline to ensure spatial uniformity and stable gradient propagation. All images and maps are spatially resized to a fixed resolution of 224×224 pixels.

To augment the training data and improve the model's robustness against varying lighting conditions, a color jitter transformation is applied to the input RGB images, introducing random variations in brightness (± 0.2), contrast (± 0.2), saturation (± 0.2), and hue (± 0.1). Subsequently, the images are converted into PyTorch tensors and standardized using the ImageNet [4] channel-wise mean ($[0.485, 0.456, 0.406]$) and standard deviation ($[0.229, 0.224, 0.225]$). The ground truth maps are solely resized and converted to single-channel tensors with continuous probability values constrained between 0.0 and 1.0.

B. Feature Extraction

Unlike traditional machine learning pipelines that rely on handcrafted feature descriptors (e.g., Gabor filters or SIFT)

computed over sliding windows, our approach leverages an end-to-end deep learning paradigm.

Explicit manual feature extraction is bypassed; instead, the pre-processed and normalized RGB pixel tensors are fed directly into the network. Feature extraction is performed automatically and hierarchically by the Swin-Transformer encoder.

C. Dataset Splitting

Following the dataset splitting criteria required for a rigorous performance evaluation, the data is partitioned into three independent subsets. We employ the complete SALICON training set, comprising 10,000 images, to optimize the network parameters.

Because the official benchmark test annotations are held out, the original validation set of 5,000 images is randomly split into two equal halves: a validation subset of 2,500 images and a test subset of 2,500 images. The validation subset is dedicated exclusively to monitoring convergence, triggering early stopping, and executing the post-processing hyperparameter grid search. The remaining test subset is strictly reserved for the final quantitative evaluations reported in Sec. VI.

V. LEARNING FRAMEWORK

To accurately capture the complex nature of human visual attention, our learning framework showed in Fig. 2 is driven by a custom Hybrid Loss function that simultaneously penalizes probability distribution divergence and maximizes spatial linear correlation.

A. Custom Hybrid Loss Formulation

Traditional saliency models often rely on a single metric for optimization, which may fail to capture both the subtle continuous distribution of human fixations and their structural peaks. To address this, we combine the Kullback-Leibler (KL) divergence and the Pearson Correlation Coefficient (CC) into a unified loss function.

Let $\mathbf{p}_i$ and $\mathbf{g}_i$ represent the flattened, 1D vectors of the predicted saliency map and the ground truth map, respectively, for the i -th image in a batch of size N . Before computing the divergence, both vectors are normalized into valid probability distributions such that their elements sum to 1. The KL divergence loss, which measures the information lost when $\mathbf{p}_i$ is used to approximate $\mathbf{g}_i$, is defined as:

$$\mathcal{L}_{KL} = \frac{1}{N} \sum_{i=1}^N \sum_j \mathbf{g}_{i,j} \log \left(\frac{\mathbf{g}_{i,j} + \epsilon}{\mathbf{p}_{i,j} + \epsilon} \right), \quad (1)$$

where $\epsilon = 10^{-7}$ is a small stabilizing constant to prevent undefined logarithmic operations.

To explicitly enforce structural and linear agreement, we incorporate a CC-based loss. Let $\text{cov}(\cdot, \cdot)$ denote the covariance and $\sigma(\cdot)$ denote the standard deviation. Since the CC metric inherently ranges from -1 to 1 (where 1 indicates

perfect correlation), we formulate the loss to minimize the discrepancy ($1 - \text{CC}$):

$$\mathcal{L}_{\text{CC}} = 1 - \frac{1}{N} \sum_{i=1}^N \frac{\text{cov}(\mathbf{p}_i, \mathbf{g}_i)}{\sigma(\mathbf{p}_i)\sigma(\mathbf{g}_i) + \epsilon}. \quad (2)$$

The total hybrid objective function is computed as the weighted sum of these two components:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{KL}} + \lambda \mathcal{L}_{\text{CC}}, \quad (3)$$

where the weighting hyper-parameter λ is empirically set to 1.0 to ensure both distribution matching and spatial correlation are given equal importance during gradient descent.

B. Optimization and Training Schedule

The network parameters are optimized using the Adam optimizer. We set a base learning rate of 1×10^{-5} combined with an L2 regularization (weight decay) penalty of 1×10^{-4} to mitigate overfitting on the training set.

Due to the stringent memory constraints imposed by the high-resolution feature maps of the Swin-Transformer backbone, we implement Gradient Accumulation to artificially increase our batch size. The forward and backward passes are executed using a micro-batch size of 2. Gradients are accumulated over 4 successive steps before performing a single optimizer step, effectively simulating a macro-batch size of 8.

The training process is scheduled for a maximum of 20 epochs. To further prevent overfitting and ensure optimal generalization, we employ an Early Stopping mechanism monitored via the validation loss. If the validation hybrid loss fails to improve for a patience parameter of 5 consecutive epochs, the training is halted, and the model weights from the best-performing epoch are restored for final inference.

VI. RESULTS

In this section, we present a comprehensive quantitative and qualitative evaluation of the proposed architecture. We begin by detailing the hyperparameter tuning conducted on the validation set, followed by the model’s evaluation on the test set under strict inference constraints. Finally, we conduct an ablation study to isolate the impact of regularization and post-processing techniques, concluding with a comparative analysis against state-of-the-art baselines.

A. Hyperparameter Optimization and Post-Processing

To optimize the quality of predicted continuous probability distributions, a post-processing Gaussian blur was evaluated over a grid search of kernel sizes $k \in \{3, 5, 7, 9\}$ and standard deviations $\sigma \in \{0.5, 1.0, 1.5, 2.0\}$ using the validation set.

As shown in Tab. 1, applying spatial smoothing significantly refines the continuous feature response. The optimal configuration was identified at a kernel size of 5×5 with $\sigma = 1.0$, achieving the peak Pearson Correlation Coefficient ($\text{CC} = 0.9058985$). This parameter pair was subsequently frozen for all final test evaluations.

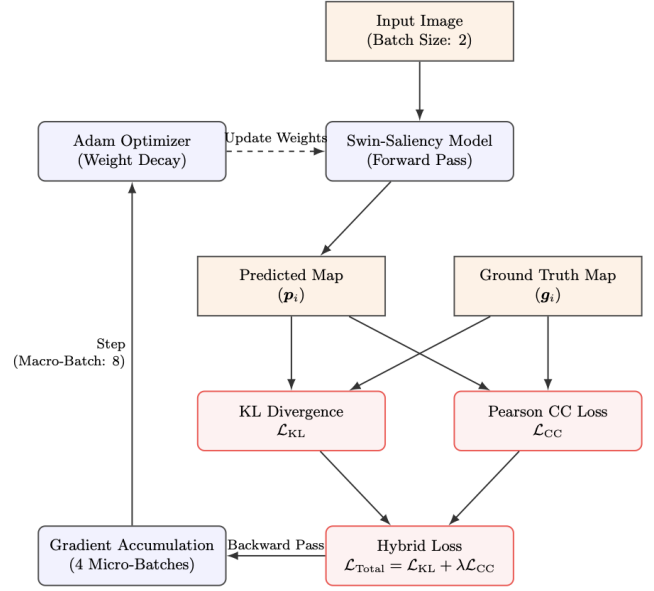


Fig. 2: Overview of the proposed learning framework. The network computes a Hybrid Loss by balancing KL divergence and Pearson CC.

TABLE 1: Grid search results for Gaussian Blur post-processing parameters on the Validation Set.

Kernel Size ($k \times k$)	Sigma (σ)	Validation CC
3×3	0.5	0.9058940
3×3	1.0	0.9058971
5×5	1.0	0.9058985
5×5	1.5	0.9058982
7×7	1.5	0.9058945
7×7	2.0	0.9058861
9×9	2.0	0.9058724

B. Quantitative Performance Evaluation

The training dynamic was monitored across 20 epochs using early stopping with a patience parameter of 5. The evolution of the hybrid loss (combining Kullback-Leibler Divergence and Pearson Correlation Coefficient) and linear correlation metrics is illustrated in Fig. 3.

To maximize the robustness of the final predictions, the model utilizes an advanced inference pipeline.

- **Multi-Scale Test-Time Augmentation (TTA):** The pipeline scales the input images by factors of 0.8, 1.0, and 1.2. For each scale, the model predicts the saliency for both the original image and a horizontally flipped version, subsequently fusing the outputs by averaging.
- **Post-Processing:** The ensemble predictions are smoothed using a Gaussian blur. The optimal hyper-parameters for the blur (kernel size and sigma) are determined dynamically via a grid-search evaluated on the validation set. The final blurred maps are clamped between 0.0 and 1.0.

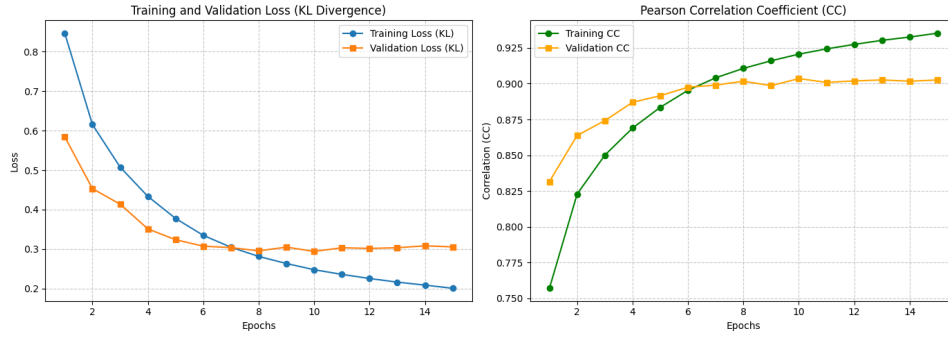


Fig. 3: Training and validation performance metrics evaluated over 15 epochs. **Left:** Evolution of the Kullback-Leibler (KL) Divergence loss. **Right:** Trajectory of the Pearson Correlation Coefficient (CC). The validation curves demonstrate rapid initial convergence, with performance plateauing in the later epochs before early stopping is triggered.

Tab. 2 presents the consolidated performance across three complementary saliency evaluation metrics: Hybrid Loss, Pearson Correlation Coefficient (CC), and Similarity (SIM).

TABLE 2: Quantitative test results on the benchmark dataset using frozen hyper-parameters.

Evaluation Metric	Test Score
Hybrid Loss ($\mathcal{L}_{KL} + \mathcal{L}_{CC}$)	0.2911
Pearson Correlation Coefficient (CC)	0.9037
Similarity (SIM)	0.7903

C. Ablation Study on Regularization and Inference

To understand the impact of the individual components within our architecture and inference pipeline, we conducted an ablation study evaluating the test set performance across three distinct configurations. As detailed in Tab. 3, the baseline model (trained without explicit regularization and evaluated without post-processing) already demonstrates solid predictive capabilities, achieving a Pearson Correlation Coefficient (CC) of 0.8985.

The introduction of spatial and parametric regularization techniques (Spatial Dropout and L2 Weight Decay) yields a marginal improvement, stabilizing the network and slightly reducing the Hybrid Loss to 0.3074 and slightly refining the similarity (SIM) score to 0.7898. However, the most significant performance leap is observed upon activating the complete inference pipeline. The integration of Multi-Scale Test-Time Augmentation (TTA) coupled with the optimized Gaussian Blur post-processing enables the model to break the 0.90 threshold, reaching a peak CC of 0.9037 and drastically lowering the total loss to 0.2911. This unequivocally demonstrates that while regularization prevents overfitting during training, it is the multi-scale aggregation during testing, combined with spatial smoothing, that guarantees the highest structural fidelity against human fixation ground truths.

D. Comparison with State-of-the-Art Baseline

To benchmark our proposed architecture, we compare our results against the Deep Visual Attention (DVA) model,

TABLE 3: Ablation study on regularization and inference.

Configuration	Loss	CC	SIM
Baseline	0.3079	0.8985	0.7891
+ Dropout & L2	0.3074	0.8992	0.7898
+ TTA & Blur	0.2911	0.9037	0.7903

a highly optimized CNN baseline utilizing a VGG16 encoder and deep supervision. As shown in Tab. 4, our Swin-Transformer model substantially outperforms the DVA framework across all evaluated datasets and metrics. This significant performance gap highlights the architectural superiority of Vision Transformers: while standard CNNs require deep convolutional stacks to expand localized receptive fields, our model intrinsically captures broad contextual dependencies and global semantics from the earliest stages of feature extraction.

TABLE 4: Quantitative comparison between the CNN-based DVA baseline and our proposed Swin-Transformer model.

Model	Dataset	CC	SIM
DVA	TORONTO	0.7200	0.5800
DVA	MIT300	0.6800	0.5800
DVA	PASCAL-S	0.6600	0.5200
Ours (Swin-T)	SALICON	0.9037	0.7903

To qualitatively demonstrate the network’s predictive capabilities, sample saliency maps generated on test stimuli are visualised in Fig. 4.

VII. CONCLUDING REMARKS

In this report, we presented an end-to-end deep learning architecture combining a Swin-Transformer encoder with a convolutional decoder, optimized via a custom Hybrid Loss, to predict human visual saliency.

The transition to a Vision Transformer was driven by empirical necessity. During our initial experiments using a standard ResNet encoder, the model’s performance plateaued, consistently failing to exceed a Pearson Correlation Coefficient of 0.89. By adopting the Swin-Transformer, we successfully captured broader global dependencies and hierarchical

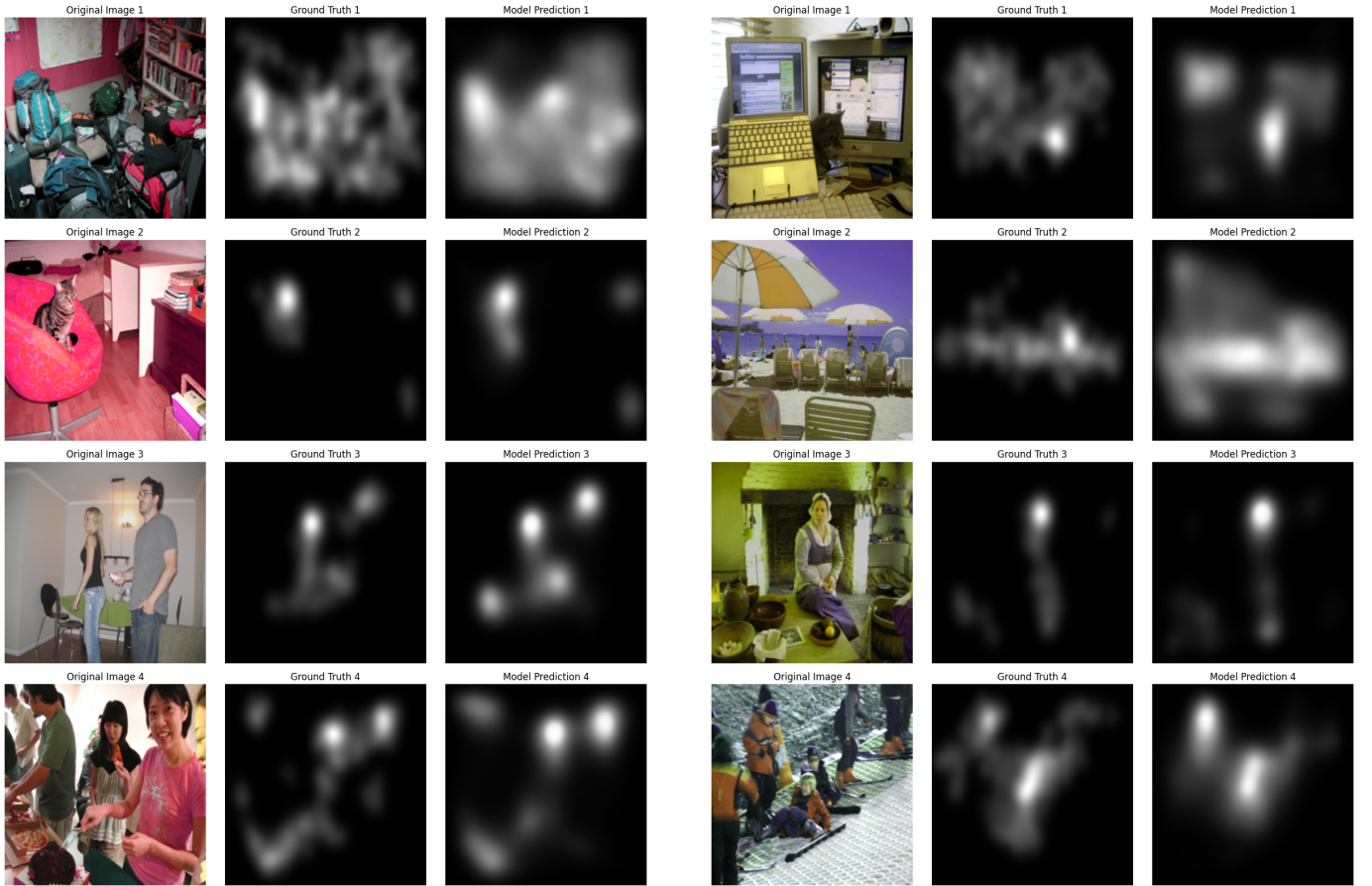


Fig. 4: Qualitative visual comparison on representative test samples. From left to right: Original Input Stimulus, Ground Truth Saliency Map, and Saliency Map predicted by the proposed model with TTA and Gaussian smoothing

features, breaking the 0.89 barrier to achieve a competitive CC of 0.9037.

Furthermore, we observed a notable shift regarding post-processing requirements. While our grid search for optimal Gaussian blur parameters provided crucial spatial refinement for the ResNet predictions, its measurable impact on the current Swin-Transformer model was minimal. This indicates that the transformer’s multi-scale representations, particularly when combined with Test-Time Augmentation, inherently produce highly smooth and well-calibrated probability distributions without relying on external spatial filtering.

While the proposed architecture achieves strong spatial agreement with human eye-tracking data, future extensions could investigate lightweight transformer variants to reduce computational overhead, or explore temporal attention mechanisms to adapt this pipeline for video saliency prediction.

Throughout this project, we gained profound practical insights into integrating modern Vision Transformers with traditional convolutional decoders. We learned how to properly permute and align multi-scale features across disparate architectures, and how to effectively balance divergent statistical metrics using a custom loss function.

The primary difficulties we encountered involved resolving spatial dimension mismatches during the skip-connection phase, managing stringent GPU memory limits during the multi-scale inference loop, and properly configuring gradient accumulation steps to stabilize the training process within our hardware constraints.

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